Seung Hyun Cha

Associate Professor, Graduate School of Culture Technology, KAIST

Associate Professor (Tenured, Sep 2021 – Present)
Graduate School of Culture Technology, KAIST

- **Director** (Sep 2021 – Present), Future Space Lab, KAIST
- **Director** (Jul 2026 – Present), FACT Research Center (Future Architecture Convergence Technology), CTRI, KAIST

AFFILIATE APPOINTMENTS

Affiliate Professor (Dec 2021 – Present), Department of Civil and Environmental Engineering, KAIST

Affiliate Professor (Sep 2022 – Present), Graduate School of Metaverse, KAIST

Affiliate Professor (May 2026 – Present), Department of AI and Future Studies, KAIST

Affiliate Professor (May 2026 – Present), Department of AX, KAIST

Affiliate Professor (Aug 2025 – Present), New York University

Member of YKAST, Young Korean Academy of Science and Technology (Dec 2022 – present)

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LINKS

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LAB INTRODUCTION VIDEO

youtu.be/4Fk_6BVnf_o

74	99	61
JOURNAL ARTICLES	CONFERENCE PAPERS	RESEARCH PROJECTS
2,978	**30**	**46**
CITATIONS	H-INDEX	I10-INDEX

Google Scholar, October 2026

SELECTED EXTERNAL SERVICE

- **Editorial Board Member**, Scientific Reports [SCIE, Q1, Top 15%], Springer Nature, 2024–
- **Associate Editor**, Journal of Computational Design and Engineering [SCIE, Q1, Top 5%], Oxford Univ. Press, 2022–
- **Associate Editor**, Architectural Science Review [A&HCI, Q1, Top 3%], Taylor & Francis, 2026–
- **Center Director**, AI Urban Research Center, Urban Design Institute of Korea, 2026–

01 Research interests

Future Cities | Human-Space Interaction | AI & Data-driven Architecture | VR/AR/Digital Twin

Architecture for Future Mobility (Robot, Autonomous Driving, Drones, UAM)

02 Academic qualifications

University of Cambridge · Cambridge, UK — November 2015

PhD in Architecture

'Dissertation title: Stochastic Space-use Prediction in light of Spatial Choice Behaviour: Modelling Space Preference of Work-related Activities'

Stanford University · Stanford, CA — June 2011

MS in Sustainable Design and Construction

Korea University · Seoul, Korea — August 2009

BEng in Architectural Engineering

03 Research and academic experience

Hanyang University · Department of Interior Architecture Design — September 2018 – August 2021

Assistant Professor — Seoul, Korea

The Hong Kong Polytechnic University · Department of Building Services Engineering — November 2015 – August 2018

Assistant Professor — Hong Kong

Nanyang Technological University · Nanyang Centre for Underground Space — March 2015 – July 2015

Research Associate — Singapore

City University of Hong Kong · Department of Architecture and Civil Engineering — December 2013 – August 2014

Graduate teaching assistant — Hong Kong

Stanford University · P5BL Lab, Department of Civil and Environmental Engineering — September 2010 - January 2011

Research assistant — Stanford, CA

Stanford University · Department of Sustainability and Energy Management — December 2009 – June 2010

Research assistant — Stanford, CA

04 Refereed journals

61 international · 13 domestic

INTERNATIONAL

61 Ma, J., Chung, D., Seol, H., Ryu, Y., and **Cha, S.** (Corresponding author), "Students' responses to lighting CCT and wall color: A cross-cultural immersive virtual reality study," Building and Environment, Elsevier, in press, 2026 [SCIE, Q1, 4.01%].

60 Seol, H., Yang, J., Jang, J., Choi, D., Han, G. (Corresponding author), and **Cha, S.**, "Development of a Physical Security Risk Matrix Integrating Heterogeneous Sensor Data and Spatiotemporal Context: Expert-Derived Parameters and Scenario-Based Validation," Architectural Research, Architectural Institute of Korea, 2026 [KCI].

59 Cho, K., Kang, J., Han, G., and **Cha, S.**(Corresponding author), "Personalizing Spatial Configuration and User Responses in Virtual Reality Workspaces," Architectural Science Review, Taylor & Francis, 1-14, 2026 [A&HCI, Q1, 0.55%].

58 **Cha, S.**, "Guest editorial: Revolutionizing workplace efficiency with VR/AR, IoT and AI: future office innovations," Journal of Corporate Real Estate, Emerald, in press, 2026 [ESCI].

57 Cho, K., Lee, S., and **Cha, S.**(Corresponding author), "Implications of User-Controlled Architectural Design on Spatial Experience in Virtual Reality Workspaces," Journal of Corporate Real Estate, Emerald, in press, 2026 [ESCI].

56 Jang, J. and **Cha, S.**(Corresponding author), "Context-integrated complex activity recognition with sector-level location encoding in residential environments," Advanced Engineering Informatics, Elsevier, 76, 105043, 2026 [SCIE, Q1, 2.0%].

55 Jung, K., Cho, K., Choi, H., Bae, S., Chun, S., and **Cha, S.**(Corresponding author), "Virtual Reality Transition to Underground: Human Responses across Sequential Spatial Transitions," Scientific Reports, Springer Nature, 16(1), 23245, 2026 [SCIE, Q1, 18%].

54 Han, G., Rhee, J., Park, A., Lee, K., and **Cha, S.**(Corresponding author), "Individual route choice and decision responses in fire evacuation scenarios using immersive virtual simulations," Developments in the Built Environment, Elsevier, 26, 100907, 2026 [SCIE, Q1, 3.2%].

53 Cho, K., Han, G., Shin, J., and **Cha, S.**(Corresponding author), "The Role of Design Personalization in Enhancing Knowledge Workers' Experience in Virtual Reality Workspaces," Virtual Reality, Springer Nature, 30(42), 2026 [SCIE, Q1, 11.2%].

52 Chun, S., Jung, K., Won, H., and **Cha, S.**(Corresponding author), "Impact of Physical Environmental Factors on Mental Health in the Hospital: A Comprehensive Review," Indoor and Built Environment, Sage, 35(2), 163-192, 2026 [SCIE, Q2, 35.3%].

51 Koo, C., Chun, S., Han, S., **Cha, S.**(Corresponding author), Lee, M., and Hong, T., "Gaps between observed and perceived indoor environmental quality: An integrated assessment in multiple university library spaces," Journal of Librarianship and Information Science, Sage, 58(3), 1583-1598, 2026 [SSCI, Q2, 35.8%].

50 Luo, Y., Hasanzadeh, S., Seo, J., and **Cha, S.**, "Unraveling neural patterns across situational awareness levels on hazard recognition behaviors: a fNIRS study," Advanced Engineering Informatics, Elsevier, 68, Part A, 103628, 2025 [SCIE, Q1, 2.0%].

49 Lee, J., Kim, Y., Shin, E., Choo, S., and **Cha, S.**(Corresponding author), "An AI-assisted Approach for Creating and Archiving Interior Design References using 360-degree Panoramic Images," Architectural Science Review, Taylor & Francis, 68(6), 506-518, 2025 [A&HCI, Q1, 0.55%].

48 Luo, Y., Abbas, A., Seo, J., Ahn, S., and **Cha, S.**, "Understanding Hazard Recognition Behaviors through Situational Awareness Assessment in Virtual Construction Environments," Journal of Management in Engineering, ASCE, 41(4), 04025017, 2025 [SCIE, Q1, 4.0%].

47 Shi, M., Seo, J., **Cha, S.**, Xiao, B., and Chi, H., "Generative AI-powered architectural exterior conceptual design based on the design intent," Journal of Computational Design and Engineering, Oxford University Press, 11(5), 125-142, 2024 [SCIE, Q1, 11.5%].

46 Lee, J., Yoo, Y., and **Cha, S.** (Corresponding author), "Generative Early Architectural Visualizations: Incorporating Architect's Style-trained Models," Journal of Computational Design and Engineering, Oxford University Press, 11(5), 40-59, 2024 [SCIE, Q1, 11.5%].

45 Lee, J., Jung, H., Kim, Y. and **Cha, S.** (Corresponding author), "Creating Spatial Visualizations using Fine-tuned Interior Design Style Models Informed by User Preferences," Advanced Engineering Informatics, Elsevier, 62, Part B, 102686, 2024 [SCIE, Q1, 2.0%].

44 Cho, K., and **Cha, S.**, Immersive Computing: Perspectives on Spatial Design Strategies for Individual Virtual Reality Workspace Platforms, Archives of Design Research, 37(3), 105-117, 2024 [Scopus].

43 Hong, S., Jang, E., Cho, J., Lee, J., Rhee, J., Lee, H., Lee, M., **Cha, S.**, Koo, C., Baik, O., and Heo, Y., A living lab to develop smart home services for the residential welfare of older adults, Technology in Society, Elsevier, 77, 102577, 2024 [SSCI, Q1, 0.5%].

42 Lee, J., **Cha, S.**, Hong, T., and Koo, C., Empirical investigation of occupant-centric thermal comfort in hotel guestrooms, Renewable and Sustainable Energy Reviews, Elsevier, 189, Part B, 114046, 2024 [SCIE, Q1, 3.3%].

41 Hwang, K., Kim, I., Kim, J., and **Cha, S.**, Client-engaged Collaborative Pre-Design Framework for Modular Housing, Automation in Construction, Elsevier, 156, 105123 2023 [SCIE, Q1, 1.1%].

40 Luo, Y., Ahn, J., Abbas, A., **Cha, S.**, Kim, J. and Seo, J., "Investigating the Impact of Scenario and Interaction Fidelity on Training Experience When Designing Immersive Virtual Reality-based Construction Safety Training," Developments in the Built Environment, Elsevier, 16, 100223, 2023 [SCIE, Q1, 3.2%].

39 Ma, J., Erdogmus, E., and Cha, S (Corresponding author), "Integration of a choice modeling approach with immersive virtual environments for accurate space utilization prediction," Journal of Building Engineering, Elsevier, 76, 107126, 2023 [SCIE, Q1, 9.0%].

38 Lee, J., Cho, K., Choi, H., Choi, S., Kim, S. and **Cha, S.** (Corresponding author), "High-level implementable methods for automated building code compliance checking," Developments in the Built Environment, Elsevier, 15, 100174, 2023 [SCIE, Q1, 3.2%].

37 Lee, J., Schermer, B., and **Cha, S.** (Corresponding author), "Effects of indoor vegetation density on human well-being for a healthy built environment," Developments in the Built Environment, Elsevier, 14, 100172, 2023 [SCIE, Q1, 3.2%].

36 **Cha, S.**, Ma, J., Seo, J., Kim, J., and Han, S., “Empirical comparison of spatial experience between photo-based IVE and real space,” Architectural Science Review, Taylor & Francis, 66(1), 1-16, 2023 [A&HCI, Q1, 0.55%].

35 Lee, J., Ma, J., Seo, J., and **Cha, S.** (Corresponding author), “Review of Applications and User Perceptions of Smart Home Technology for Health and Environmental Monitoring,” Journal of Computational Design and Engineering, Oxford University Press, 9(3), 857–889, 2022 [SCIE, Q1, 9.44%].

34 Ma, J., Lee, J. and **Cha, S.** (Corresponding author), "Effects of lighting CCT and illuminance on visual perception and task performance in immersive virtual environments," Building and Environment, Elsevier, 209, 108678, 2022 [SCIE, Q1, 4.01%]

33 Gassar, A., and **Cha, S.** (Corresponding author), " Feasibility assessment of adopting distributed solar photovoltaics and phase change materials in multifamily residential buildings,” Sustainable Production and Consumption, Elsevier, 29, 507-528, 2022 [SSCI, Q1, 23.6%].

32 Jo, T., Ma, J. and **Cha, S.** (Corresponding author), "Elderly Perception on the Internet of things-based Integrated Smart-Home System," Sensors, MDPI, 21(4), 1284, 2021 [SCIE, Q1, 21.09%].

31 Gassar, A., and **Cha, S.** (Corresponding author), "Review of geographic information systems-based rooftop solar photovoltaic potential estimation approaches at urban scales,” Applied Energy, Elsevier, 291, 116817, 2021 [SCIE, Q1, 3.85%].

30 Ma, J. and **Cha, S.** (Corresponding author), "A user-specific activity pattern generation framework for evidence-based ABW planning," Building and Environment, Elsevier, 189, 107519, 2021 [SCIE, Q1, 4.01%]

29 Jang, J., Ahn, S., **Cha, S.**, Cho, K., Koo, C. and Kim, T., “Toward Productivity in Future Construction: Mapping Knowledge and Finding Insights for Achieving Successful Offsite Construction Projects,” Journal of Computational Design and Engineering, Oxford University Press, 8(1), 1-14, 2021 [SCIE, Q1, 9.44%].

28 Gassar, A., Koo, C., Kim, T. and **Cha, S.** (Corresponding author), " Performance Optimization Studies on Heating, Cooling and Lighting Energy Systems of Buildings during the Design Stage: A Review,” Sustainability, MDPI, 13, 9815, 2021 [SCIE, Q2, 45.07%].

27 Ma, J. and **Cha, S.** (Corresponding author), “A human data-driven interaction estimation using IoT sensors for workplace design,” Automation in Construction, Elsevier, 119, 103352, 2020 [SCIE, Q1, 1.09%].

26 **Cha, S.**, Zhang, S., and Kim, T., “Effects of Interior Color Schemes on Emotion, Task Performance, and Heart Rate in Immersive Virtual Environments,” Journal of Interior Design, Wiley, 45(4), 51-65, 2020 [A&HCI, Q1, 8.24%].

25 Gassar, A., and **Cha, S.** (Corresponding author), "Energy prediction techniques for large-scale buildings towards a sustainable built environment: A review,” Energy and Buildings, Elsevier, 224, 110238, 2020 [SCI, Q1, 6.2%].

24 Li, W., Koo, C., Hong, T., Oh, J., **Cha, S.** and Wang, S., "A novel operation approach for the energy efficiency improvement of the HVAC system in office spaces through real-time big data analytics," Renewable and Sustainable Energy Reviews, Elsevier, 127, 109885, 2020 [SCIE, Q1, 1.14%].

23 **Cha, S.** and Kim, T., "The role of space attributes in space-choice behaviour and satisfaction in an academic library," Journal of Librarianship and Information Science, Sage, 52(2), 399-409, 2020 [SSCI, Q3, 54.71%].

22 Abbas, A., Choi, M., Seo, J., **Cha, S.**, and Li, H., "Effectiveness of Immersive Virtual Reality-based Communication for Construction Projects," KSCE Journal of Civil Engineering, KSCE, 23(12), 4972-4983, 2019 [SCIE, Q3, 68.98%].

21 **Cha, S.**, Koo, C., Kim, T., and Hong, T., "Spatial perception of ceiling height and type variation in immersive virtual environments," Building and Environment, Elsevier, 163, 106285, 2019 [SCI, Q1, 4.01%].

20 Kim, T. and **Cha, S.** (Corresponding author), "Empirical validation of the spatial-choice modeling approach to user simulation," Architectural Science Review, Taylor & Francis, 62(4), 313-322, 2019 [A&HCI, Q1, 0.55%].

19 Chen, H., **Cha, S.**, and Kim, T., "A framework for group activity detection and recognition using smartphone sensors and beacons," Building and Environment, Elsevier, 158, 205-216, 2019 [SCI, Q1, 4.01%].

18 Zhang, S., Lu, L., Dong, C., and **Cha, S.**, "Performance evaluation of a double-pipe heat exchanger fitted with self-rotating twisted tapes," Applied Thermal Engineering, Elsevier, 158, 113770, 2019 [SCIE, Q1, 9.17%].

17 Baek, S., and **Cha, S.** (Corresponding author), "The trilateration-based BLE Beacon system for analyzing user-identified space usage of New Ways of Working offices," Building and Environment, Elsevier, 149, 264-274, 2019 [SCI, Q1, 4.01%].

16 LI, W., Koo, C., **Cha, S.**, Lai, H., and Lee, J., "A conceptual framework for the real-time monitoring and diagnostic system for the optimal operation of smart building: A case study in Hotel ICON of Hong Kong, " Energy Procedia, Elsevier, 158, 3107-3112, 2019 [Scopus , Q1, 23.08%].

15 Koo, C., Li, W., **Cha, S.** (Corresponding author), Zhang, S., "A novel estimation approach for the solar radiation potential with its complex spatial pattern via machine-learning techniques," Renewable Energy, Elsevier, 133, 575-592, 2019 [SCIE, Q1, 13.60%].

14 Zhang, S., Lu, L., Dong, C., and **Cha, S.**, "Thermal characteristics of perforated self-rotating twisted tapes in a double-pipe heat exchanger," Applied Thermal Engineering, Elsevier, 162, 114296, 2019 [SCIE, Q1, 9.17%].

13 Li, W., Koo, C., **Cha, S.**, Hong, T., and Oh, J., "A novel real-time method for HVAC system operation to improve indoor environmental quality in meeting rooms," Building and Environment, Elsevier, 144, 365-385, 2018 [SCI, Q1, 4.01%].

12 **Cha, S.**, Seo J., Baek, S., and Koo, C., "Towards a well-planned, activity-based work environment: Automated recognition of office activities using accelerometers," Building and Environment, Elsevier, 144, 86-93, 2018 [SCI, Q1, 4.01%].

11 **Cha, S.**, Steemers, K., and Kim, T., "Modeling space preferences for accurate occupancy prediction during the design phase," Automation in Construction, Elsevier, 93, 135-147, 2018 [SCIE, Q1, 1.09%].

10 Oh, J., Koo, C., Hong, T. and **Cha, S.**, "An integrated model for estimating the techno-economic performance of the distributed solar generation system on building façades: Focused on energy demand and supply," Applied Energy, Elsevier, 228, 1071-1090, 2018 [SCI, Q1, 3.85%].

9 Kim, J., **Cha, S.** (Corresponding author), Koo, C. and Tang, S., "The effects of indoor plants and artificial windows in an underground environment," Building and Environment, Elsevier, 138, 53-62, 2018 [SCI, Q1, 4.01%].

8 Kim, T., **Cha, S.** and Kim, Y., "Space choice, rejection, and satisfaction in university campus," Indoor and Built Environment, Sage, 27 (2), 233-243, 2018 [SCIE, Q2, 42.12%]

7 **Cha, S.**, Steemers, K., and Kim, T., "Modelling building users' space preferences for group work: A discrete choice experiment," Architectural Science Review, Taylor & Francis, 60(6), 460-471, 2017 [A&HCI, Q1, 0.55%].

6 Koo, C., Jin, R., Li, B., **Cha, S.**, and Wanatowski, D., "Case-based reasoning approach to estimating the strength of sustainable Concrete," Computers and Concrete. Techno-press, 20 (6), 645-654, 2017 [SCIE, Q1, 20.07%].

5 Kim, T., **Cha, S.** and Kim, Y., "A framework for evaluating user involvement methods in architectural, engineering, and construction projects," Architectural Science Review, Taylor & Francis, 59(2), 136-147, 2016 [A&HCI, Q1, 0.55%].

4 Kim, T., Kim, Y. **Cha, S.** (Corresponding author) and Fischer, M., "Automated updating of space design requirements connecting user activities and space types," Automation in Construction, Elsevier, 50, 102-110, 2015 [SCIE, Q1, 1.09%].

3 **Cha, S.** and Kim, T., "What matters for students' use of physical library space?," Journal of Academic Librarianship, Elsevier, 41(3), 274-279, 2015 [SSCI , Q3, 61.76%].

2 **Cha, S.** and Kim, T., "A conceptual framework of agent-based space-use prediction simulation system," Journal of Construction Engineering and Project Management, Korea Institute of Construction Engineering and Management, 5(4), 12-15, 2015.

1 **Cha, S.** and Kim, T., "A conceptual framework for developing indoor spatial choice model," International Journal of Engineering and Technology, IACSIT Press, 6(5), 386-391, 2014.

DOMESTIC

13 Kang, J., Ryu, Y., Ji, J., Eum, Y., and **Cha, S.** (Corresponding author), "A Framework for Adaptive Information Presentation in Conversational Navigation Considering Age-Related Wayfinding Characteristics," Journal of the Korean Institute of Spatial Design, 2026 [KCI].

12 Seol, H., Choi, H., Kang, J., and **Cha, S.** (Corresponding author), "A Queryable Scene Representation Based on Object-Light Contribution Relations for Lighting Design Decision Support in Digital 3D Spaces," Journal of the Korean Institute of Spatial Design, 2026 [KCI].

11 Bae, S., Bae, S., Kim, H., and **Cha, S.** (Corresponding author), “A Framework for XR Museums in Idle School Classrooms: An Implementation Strategy to Mitigate Cultural Accessibility Gaps,” Journal of Digital Contents Society, 2026 [KCI].

10 Kim, D., Eum, Y., Yoon, B., Lee, H., Kim, S., Lim, Y., Baeck, M., Kim, M., Jung, D., Jeong, S., **Cha, S.**, Kim, S., and Woo, W., “An Intelligent XR Platform for Meta-Object Creation/Management and Real-Time Gesture-Responsive Multimodal Feedback,” Journal of KIISE, 53(6), 476-488, 2026 [KCI].

9 Han, G., Song, Y., Han, J., and **Cha, S.** (Corresponding author), “Development of Level-of-Service (LOS) Indicators for Evaluating Human–Robot Coexistence Performance in Robot-Friendly Built Environments,” Journal of the Architectural Institute of Korea, 42(5), 71-82, 2026 [KCI, SCOPUS].

8 Song, Y., Jang, J. and **Cha, S.** (Corresponding author), “Development of an Automated Office Room Generation Methodology Based on Space Utilization,” Journal of KIBIM, 14(3), 1-10, 2024 [KCI].

7 Chun, S. and **Cha, S.** (Corresponding author), “A Study on Spatial Experience Using Virtual Reality for Efficient Architectural Design,” Journal of KIBIM, 14(1), 1-10, 2024 [KCI].

6 Chun, S. and **Cha, S.** (Corresponding author), “A Study on the Applicability of Metaverse Experience and Technology to the Architectural Design Process,” Journal of KIBIM, Korean Institute of Building Information Modeling, 13(2), 16-28, 2023 [KCI].

5 Rhee, J. and **Cha, S.** (Corresponding author), “A Study on the Meaning, Types, and Characteristics of Housing in the Metaverse,” LHI Journal, 13(4), 91-103, 2022.

4 Cho, K., Eum, Y., Choi, H., Lee, S., and **Cha, S.** (Corresponding author), “An Analysis of Current Virtual Workspace Design and Suggestions for Future Design Methods: A Comparative Analysis of 12 Current Cases ,” Journal of KIBIM, 12(4), 80-92, 2022 [KCI].

3 Rhee, J. and **Cha, S.** (Corresponding author), “A Review of Metaverse Architectural Design through Case Studies,” Journal of the Architectural Institute of Korea, 38(7), 69-80, 2022 [KCI, SCOPUS].

2 Rhee, J. and **Cha, S.** (Corresponding author), “An Empirical Study on Long-Term Monitoring of Smart Home Systems for Older Adults,” Journal of KIBIM, Korean Institute of Building Information Modeling, 11(3), 75-90, 2021 [KCI].

1 Ma, J. and **Cha, S.** (Corresponding author), “BLE Beacon-Based Estimation of Space-Use Information for NWW Office Planning,” Korean Journal of Computational Design and Engineering, 25(3), 227–235, 2020 [KCI].

05 Books and book chapters

BOOKS

- Cha, S., *Comprehensive Introduction to Future Architectural Design: Data, Artificial Intelligence, and New User Agents* (미래건축설계총론: 데이터, 인공지능, 새로운 이용주체), Sigongmunhwasa, 2026

BOOK CHAPTERS

- "What Kind of House Will I Live in in the Metaverse? Metaverse and Architectural Design," Post Metaverse, Porche, 2021

06 Refereed conferences

51 international · 48 domestic

INTERNATIONAL

51 Ji, J., and **Cha, S.**, "Diagnosing Walking Resistance in Age-Friendly Seoul with GeoAI", The 15th International Symposium on Architectural Interchanges in Asia (ISAIA), Daejeon, Korea, 19-22 October 2026.

50 Ryu, N., Choi, H., Lee, S., and **Cha, S.**, "Generating Shopper Feedback for Pop-Up Store Planning", The 15th International Symposium on Architectural Interchanges in Asia (ISAIA), Daejeon, Korea, 19-22 October 2026.

49 Yang, J., and **Cha, S.**, "Smart Building-Mediated Mobility Hub (SBMMH): A Coordination Framework", The 15th International Symposium on Architectural Interchanges in Asia (ISAIA), Daejeon, Korea, 19-22 October 2026.

48 Lee, S., Ryu, N., Kang, J., and **Cha, S.**, "Levels of Detail in BIM-based Egocentric Spatial Perception for Generative Agent Wayfinding", The International Conference of Sociedad Iberoamericana de Gráfica Digital (SIGraDi), Santiago, Chile, 18-20 November, 2026.

47 Choo, S., Kim, Y., and **Cha, S.**, "Interaction as an Adaptive Design Variable in AI-Mediated XR Environments", The International Conference of Sociedad Iberoamericana de Gráfica Digital (SIGraDi), Santiago, Chile, 18-20 November, 2026.

46 Ji, J., Choi, H., and **Cha, S.**, "Revealing Hidden Walking Burdens: A GeoAI-Based Micro-Scale Spatial Resistance Framework for Age-Friendly Urban Communities", The International Conference of Sociedad Iberoamericana de Gráfica Digital (SIGraDi), Santiago, Chile, 18-20 November, 2026.

45 Kim, Y., Choo, S., Eum, Y., Jeon, D., and **Cha, S.**, "Designing an Augmented Reality-Based Explanation Interface for Understanding Multi-Device Automation Changes in Smart Homes", The International Conference of Sociedad Iberoamericana de Gráfica Digital (SIGraDi), Santiago, Chile, 18-20 November, 2026.

44 Cho, K., Kang, J., Eum, Y., and **Cha, S.**, "How Unreal Should a Workspace Be? A Conceptual Framework for Unrealistic Spatial Design in Virtual Reality Workspaces", The 5th Transdisciplinary Workplace Research Conference, Zurich, Switzerland, 2-5 September 2026.

43 Jeon, D., Eum, Y., Kim, Y., and **Cha, S.**, "Designing Reality-to-Virtual Transitions: Cognitive Mechanisms Beyond Visual Form", Design Computing and Cognition'26, Paris, France, 8–10 July 2026.

42 Chung,D., Bae, S., Han, G., and **Cha,S.**, "The Effect of Artwork Size on Visitor Attention in Museum Curatorial Arrangement", Design Computing and Cognition'26, Paris, France, 8–10 July 2026.

41 Han, G., Cho, K., Eum, Y., Song, Y., Park, A., and **Cha, S.**(Corresponding author), “Curriculum-based Reinforcement Learning for Fire Evacuation Simulation in Complex Indoor Environments”, Proceedings of the 2026 Annual Simulation Conference (ANNSIM’26), Society for Modeling & Simulation International (SCS), Orlando, FL, USA, 4-7 May 2026.

40 Kang, J., Chung, D., Cho, K., and **Cha,S.**, “Strategic Design Framework for Surrealistic Spaces in Virtual Reality”, Proceedings of the 12th International Conference on Digital and Interactive Arts (ARTECH 2025), Braga, Portugal, 26-28 November 2025.

39 Choi, H., and **Cha,S.**, “Processed Experience: A Contemporary Discourse on Existence with Spatial Installation", Proceedings of the 12th International Conference on Digital and Interactive Arts (ARTECH 2025), Braga, Portugal, 26-28 November 2025.

38 Shin, J., Chung, D., Bae, S., and **Cha, S.**, “A framework for analyzing XR media art through impossible spaces and interaction structures”, Proceedings of the 12th International Conference on Digital and Interactive Arts (ARTECH 2025), Braga, Portugal, 26-28 November 2025.

37 Choi, H., and **Cha,S.**, “Quivering Moment", Proceedings of the 22nd annual International Conference Architectural Humanities Research Association (AHRA 2025), Liverpool, UK, 22-24 November 2025.

36 Cho, K., Kang, J., and **Cha,S.**, “Beyond Physical Boundaries: The Concept and Characteristics of Cognitive Spatial Segmentation”, Proceedings of Asia Design and Innovation Conference, Shanghai, China, 20-21 November 2025.

35 Choi, H., and **Cha,S.**, “Egocentric vision data for confined space design: Opportunities, challenges, and strategies”, Proceedings of Asia Design and Innovation Conference, Shanghai, China, 20-21 November 2025.

34 Kang, J., Eum, Y., Cho, K., Seol, H., and **Cha,S.**, “Integrating Human, Space, and Kinematics in Interactive Architecture for Experimental Framework”, Proceedings of the 11th International Conference on Smart and Sustainable Built Environment (SASBE 2025), Lille, France, 3-5 November 2025.

33 Choi, H., Choo, S., and **Cha,S.**, “Design Guideline for Virtual Indoor Gardening in Extended Reality”, Proceedings of the 11th International Conference on Smart and Sustainable Built Environment (SASBE 2025), Lille, France, 3-5 November 2025.

32 Choi, H., Shin, J., Choo, S., and **Cha,S.**, “Gesture-based Survey Design for Evaluating Indoor Spatial Experience in Extended Reality”, Proceedings of the 11th International Conference on Smart and Sustainable Built Environment (SASBE 2025), Lille, France, 3-5 November 2025.

31 Cho, K., Kang, J., Jeon, D., and **Cha,S.**, “Seamless Transitional Spaces for Sustainable Integration in Immersive Virtual Reality”, in Proceedings of the 11th International Conference on Smart and Sustainable Built Environment (SASBE2025), Lille, France, 3-5 November 2025.

30 Bae, S., Eum, Y., Kim, H., Kim, Y., Jeon, D., Jang, J. and **Cha,S.**, “Adaptive Virtual Reality Space Design Based on Real-Time User Emotion”, 2025 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct), Daejeon, Korea, 8-12 October, 2025.

29 Choi, H., Lee, J., Shin, J., and **Cha, S.**, “A Framework for Virtual Windows to Enhance Indoor Spatial Experience in Extended Reality,” The International Conference of Computer-Aided Architectural Design Futures (CAAD Futures 2025), Hong Kong, 2-4 July, 2025.

28 Choi, H. and **Cha, S.**, “Projectional Dialogue: A Contemporary Discourse on Dialogue with AR-based Spatial Installation", International Symposium on Electronic/Emerging Art (ISEA 2025), Seoul, Korea, 23-29 May, 2025

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DOMESTIC

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47 Song, Y., Kim, B. and **Cha, S.**, “A Preliminary Study on Information Utilization by Design Task in BIM-Based AI Agents for Architectural Design”, 2026 Fall Conference of AIK (Architectural Institute of Korea), Korea, 2026.

46 Song, Y. and **Cha, S.**, “A Preliminary Study of Shared Circulation Spaces and Human-Robot Mobility in Robot-Friendly Buildings”, 2026 Fall Conference of AIK (Architectural Institute of Korea), Korea, 2026.

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40 Kim, Y., Eum, Y., Jeon, D., Bae, S., and **Cha, S.**, “Designing an Augmented Reality-Based Explanation Interface for Understanding Multi-Device Automation Changes in Smart Homes,” Summer Conference of Society for Computational Design and Engineering, Jeju, Korea, 19-22 August, 2026.

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36 Cho, K., and **Cha, S.**, “Design Personalization and User Stress in Virtual Reality Workspaces: A Conceptual Model for Biosignal-based Validation,” Summer Conference of Society for Computational Design and Engineering, Jeju, Korea, 19-22 August, 2026.

35 Jeon, D., Kim, Y., Bae, S., and **Cha, S.**, “AI-Based Personalized Visual Cue System for Supporting Attention Recovery in VR Meditation for Novice Meditators,” Summer Conference of Society for Computational Design and Engineering, Jeju, Korea, 19-22 August, 2026.

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33 Seol, H. and **Cha, S.**,, “A Case Study on Lighting Placement Automation by Space Type - The Need for Research on Context-aware Lighting Automation in Digital Dynamic Spaces,” KISD Fall International Conference, Seoul, Korea, 01 November, 2025

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30 Shin, J., Chio, H. and **Cha, S.**, “Visualization Methodology for Pedestrian Heterogeneity in Agent-based Simulation,” 80th Anniversary Academic Presentation Contest of AIK (Architecture Institute of Korea), Seoul, Korea, 29-31 October, 2025.

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22 Kim, H., Bae, S. and **Cha, S.**, “Proposal for Design Research Directions to Enhance Virtual Spatial Experience - Focused on Visual Perception Research,” Winter Conference of Society for Computational Design and Engineering annual conference, Pyeongchang, Korea, 29-31 January, 2024.

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20 Lee, J., Song, Y. and **Cha, S.**, “An Investigation of Spatial Perception in Virtual Reality – Through the Exploration of Peripersonal Space,” Winter Conference of Society for Computational Design and Engineering annual conference, Pyeongchang, Korea, 29-31 January, 2024.

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15 Jang, J., Song, Y., and **Cha, S.**, “Automatic floorplan generation considering energy efficiency - Focused on the Energy Performance Prediction,” The Annual Autumn Conference of AIK (Architecture Institute of Korea), Jeongseon, Korea, 25-27 October, 2023.

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12 Song, Y., and **Cha, S.**, “A study of the Automation of Office Space Layout Generation: Focused on the Space Utilization Prediction,” The Annual Spring Conference of AIK (Architecture Institute of Korea), Busan, Korea, 26-28 April, 2023.

11 Choi, H., Cho, K., and **Cha, S.**, “A Proposal for Designing a Living Space for Using Virtual Reality and Augmented Reality: The Concept of Transitional Space Its Applicability,” The Annual Spring Conference of AIK (Architecture Institute of Korea), Busan, Korea, 26-28 April, 2023.

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5 Jo, T., Ma, J., and **Cha, S.**, Proposal for an Integrated IoT-based Smart Home Environment to Monitor the Healthcare of Older Adults for Aging in Place, Summer Conference of Society for Computational Design and Engineering annual conference, Seoul, Korea, 17 Aug, 2020.

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3 **Cha, S.**, IoT-based indoor positioning and activity recognition system, KIBIM annual conference, Seoul, Korea, 13 June, 2019.

2 Lim, J. and **Cha, S.**, The Role of Tactile and Olfactory Sense in Design Review Using Virtual Reality, KIBIM annual conference, Seoul, Korea, 13 June, 2019.

1 Lee, J., Koo, C., and **Cha, S.**, A Real-Time Management Framework for the Energy Efficiency in a Room of Building via IoT-based Sensor Network: Focused on Hotel Guestroom, Korea Institute of Construction Engineering and Management (KICEM) annual conference, Seoul, Korea, 9 November, 2018.

07 Non-refereed publications

- Lee, J. and **Cha. S.**, “The Potential of the Metaverse in the Architecture and Construction Industry,” The BIM Magazine, 25, 23-27, buildingSMART Korea, 2022
- Lee, J. and **Cha. S.**, “Green Wall Techniques and Application Cases,” Korean Institute of Architectural Sustainable Environment and Building Systems, 15(2), 2021
- Ma, J. and **Cha. S.**, “Research Trend of Space Use Data Estimation for Applying NWW Office Planning,” Korea Institute of Building Information Modelling (KIBIM), 9(2), 16-22, 2019
- Baek, S. and **Cha, S.**, “Research Trends in measuring occupant behavior towards for the application of activity-based working environment,” Construction Engineering and Management, Korea Institute of Construction Engineering and Management (KICEM), 18(6), 53-55, 2017.
- **Cha, S.** and Kim, T., “Global Research Trends in Prediction of Occupancy schedules for Building Simulations,” Construction Engineering and Management, Korea Institute of Construction Engineering and Management (KICEM), 15(5), 52-54, 2014.

08 Patents

- Bae, S., Eum, Y., Jang, J., Kim, Y., and Jeon, D., "System and Method for Providing Emotion-Based Adaptive Spatial Content," Korean Patent Application No. 10-2025-0143692 (Applied 2025).
- Kim, Y., Lee, H., Noh, S., **Cha, S.**, Choi, H., and Cho, K., "Virtual Underground Space Navigation Environment System," Korean Patent No. 10-2823126 (Registered: June 17, 2025).

09 Exhibitions

- Kim, K. and **Cha, S.**, “Seasonal Atrium,” 2020 International Invitation Exhibition, Korean Institute of Interior Design, Nexus Seoul Gallery, 7 November, 2020.

10 Research projects (Ongoing)

5 projects

- Space and Behavior, **PI**, Daewoo Foundation, 2027-28 (KRW20,000,000).
- AI-Based Responsive–Transformable Space Technologies for Future Space Innovation, **PI**, National Research Foundation, 2025-28 (KRW300,000,000).
- Core Technologies for the Design, Construction, Operation, and Management of Robot-Friendly Buildings, **Co-I**, Korea Agency for Infrastructure Technology Advancement, 2025-29 (KRW18,000,000,000).
- Attention-Model-Based Spatial Design Optimization Algorithms for Virtual Meditation Environments, **PI**, KAIST Meditation Science Research Institute, 2025-27 (KRW50,000,000).
- Development of a Multisensory VR-Based Pre-Construction Space Evaluation System, **PI**, National Research Foundation, 2023-28 (KRW600,000,000).

11 Researchers' Grants (Ongoing)

17 grants

- Development of Egocentric Behavioral Data Embedding System for Embodied Agent-Based Building Simulation Using Eyewear Sensors, National Research Foundation, 2026-28 (KRW50,000,000).
- Development of a BIM-based Multimodal Agent Framework for Automated Office Floor Plan Redesign, National Research Foundation, 2026-28 (KRW50,000,000).
- Research on a Spatiotemporal Synchronization-Based Intelligent Operation Framework for UAM-Ground Mobility Transfers, National Research Foundation, 2026-27 (KRW20,000,000).
- A Study on Spatial Exploration Patterns According to the Level of Entry Agency in VR, National Research Foundation, 2026-27 (KRW20,000,000).
- The Impact of Text-Input Structuring Methods in Generative AI-Based Lighting Design on the Creative Experience and Intent Fidelity, National Research Foundation, 2026-27 (KRW20,000,000).
- Commercialization Support for AI-Based Next-Generation Security Products (Type 1-2): AI-Based Integrated Physical Security Control System, Korea Internet and Security Agency, 2026 (KRW280,000,000)
- Complex Human Activity Recognition in Residential Spaces Based on Generative Sector-Level Space Segmentation and Multimodal Sensor Fusion, National Research Foundation, 2025-27 (KRW50,000,000).
- Generating Occupant-Specific Spatial Behavior Data with LoRA-Adapted Large Language Models, National Research Foundation, 2025-27 (KRW50,000,000).

- A Generative-Algorithm-Based Optimization Process for Virtual–Physical Integrated Architectural Design, National Research Foundation, 2025-28 (KRW300,000,000).
- An Age-Friendly Voice-Based VR Interface for 3D Object Manipulation Using LLMs, National Research Foundation, 2025-26 (KRW12,000,000).
- Interaction Sequence Analysis of XR-Based Memory Visualization Interfaces for Autobiographical Continuity and Companion Presence, National Research Foundation, 2025-26 (KRW12,000,000).
- An Exhibition Space Design Evaluation Framework through Quantification of Human–Space Responses, National Research Foundation, 2025-26 (KRW12,000,000).
- Human-Centered Spatial Design through Responsive Spatial Content, National Research Foundation, 2025-27 (KRW40,000,000).
- Automated Floor Plan Generation Based on Space Utilization and Satisfaction Prediction Using Generative AI, National Research Foundation, 2024-26 (KRW50,000,000).
- Crowd Density Safety in Multi-Use Facilities Based on Crowd Behavior and Spatial Environment, National Research Foundation, 2024-29 (KRW200,000,000).
- Development of an AI-Based XR Digital Twin Integrated Evacuation Simulator, National Research Foundation, 2024-29 (KRW500,000,000).
- A Design Process for Architectural Designers' Collaboration Using the Metaverse, National Research Foundation, 2022-27 (KRW200,000,000).

12 Research projects (Completed)

39 projects

- Development of Architectural Technologies for an AI Smart Building Testbed Platform, **PI**, Korea Agency for Infrastructure Technology Advancement, 2025-26 (KRW80,000,000).
- Future Office Space Planning Theory, **PI**, National Research Foundation, 2024-26 (KRW20,000,000).
- Comprehensive Study on Future Product Design Vision and Strategy for Samsung C&T Raemian, **Co-I**, Samsung C&T, 2025-26 (KRW100,000,000).
- NCP Activities for Building an International VR Knowledge Asset Archive for Global Cultural Welfare, **PI**, National Research Foundation, 2025-26 (KRW25,000,000).
- Building an International Standard Architectural Design Network Based on Generative Agent Simulation, **PI**, National Research Foundation, 2024-25 (KRW20,000,000).
- Development of AI-Based Architectural Design Automation Technologies, **Co-I**, Korea Agency for Infrastructure Technology Advancement, 2021-25 (KRW279,656,000).
- AR–AI–Cognition–Architecture Convergence Research for Trans-Spatiotemporal Exchange, **Co-I**, National Research Foundation, 2024-28 (KRW1,800,000,000).
- A Metaverse Architectural Space Design Framework, **PI**, National Research Foundation, 2022-25 (KRW60,000,000).
- Spatial Media Experience Research, **PI**, Samsung Electronics industry–academia collaboration, 2023-25 (KRW200,000,000)

- Building a UX-Based Metaverse Workspace Design Framework, National Research Foundation, 2023-25 (KRW40,000,000).
- The Effect of Green View Index on Virtual Work Experience, National Research Foundation, 2024-25 (KRW10,000,000).
- Environment-Adaptive Real-Time XR Interface Technologies, **Co-I**, Institute for Information and Communications Technology Planning and Evaluation(IITP), 2024-26 (KRW4,700,000,000).
- Optimal Intelligent Space Design Using IoT/VR/AR, **PI**, Korea Advanced Institute of Science and Technology, 2021-24 (KRW120,000,000).
- Space Design Technologies for Long-Term Human Habitability in Underground Cities Combining VR/AR and Spatial Design, **PI**, Ministry of Science and ICT, 2022-23 (KRW175,000,000).
- Preliminary Study on Future Urban Space Transformation, **PI**, LG Business Research (LG Space Institute), 2023 (KRW4,000,000).
- LG U+ Metaverse Virtual Office UX/UI Planning, LG U+, 2022 (KRW20,000,000).
- An IoT-Sensor and AI-Based Automated Activity Pattern Generation System for Data-Driven Office Planning, **PI**, National Research Foundation, 2020-23 (KRW137,500,000).
- Building a Virtual National Library and Its Service Operation Strategies, **PI**, National Library of Korea, Ministry of Culture, Sports and Tourism, 2022 (KRW92,000,000).
- Metaverse Virtual Office Space Design for Creative Collaboration, **PI**, Creative Challenge Program, 2022 (KRW20,000,000).
- Integrated Residential Welfare Services for Older Adults through an Extended Living Lab, **Co-I**, National Research Foundation, 2019-22 (KRW450,000,000).
- Space Preference Modeling Based on Immersive Virtual Environments and Choice Modeling Theory, **PI**, National Research Foundation, 2019-22 (KRW60,000,000).
- An Online VR Exhibition Platform for Revitalizing the Arts Ecosystem, **PI**, Korea Arts Management Service, 2021-22 (KRW49,000,000).
- Using Immersive Virtual Environments for User Spatial Experience Evaluation, **PI**, Hanyang University, 2021-22 (KRW15,000,000).
- Development of an AI-Based Activity Recognition Framework for Optimal Space Planning, **PI**, Hanyang University 2020-22 (KRW52,000,000).
- A Big-Data-Based Personalized Online VR Gallery Platform, 'Gallery VAir', **PI**, Ministry of SMEs and Startups, 2020-21 (KRW48,000,000).
- VR Spatial Experience Incorporating Tactile and Olfactory Senses in the Design Review Stage, **PI**, Hanyang University, 2020-21 (KRW15,000,000).
- Development of an IoT-Based Measurement and Prediction System of Occupant Behavior Data for Smart Offices, PI, National Research Foundation, 2018-20 (KRW60,000,000).
- Exploring End-User Participation in Architectural Design Using HMD-Based Immersive Virtual Environments, **PI**, Hanyang University, 2018-19 (KRW20,000,000).

- Interdisciplinary and Collaborative Learning through a BIM-based Immersive Virtual Environment in AEC Education, **External Co-I**, University Grants Committee, 2017-19, (HK$ 492,600).
- The Effectiveness of Interactive 3D Virtual Environments for Pre-Experience of Hotel Rooms, **External Co-I**, Hotel ICON R&D, 2017-19, (HK$ 49,749).
- Implementation of Food Waste Collection System for Catering Services, **External Co-I**, Recycling Fund, Hong Kong Productivity Council, 2018 – 2020, (HK$ 2,709,050).
- A Real-Time Monitoring and Diagnostics for energy efficiency and indoor environmental quality in Hotel ICON, **External Co-I**, Hotel ICON R&D, 2018-19, (HK$ 50,000).
- Integration of discrete-choice modeling theory with immersive virtual environments for modeling space preferences toward more accurate space utilization prediction, **PI**, University Grants Committee, 2018-20, (HK$466,667).
- Development of Internet of Things (IoT) - based Multi-Function Smart Window by Integrating Solar PV Blind and Active Ventilation System toward Zero Energy Building, **External Co-I**, Ministry of Land, Infrastructure and Transport (South Korea), 2017-18, (KRW 70,000,000).
- Understanding Activity-Based Working: A Framework for Integrating Indoor Positioning System and Automated Office Activity Recognition, **PI**, Internal grant, The Hong Kong Polytechnic University, 2017-19 (HK$105,000).
- An Experimental Investigation of the Energy Saving Potential by Implementing the Real-Time Sensor and Actuator: Focused on the Indoor Environmental Quality and Smart Thermostat, **Co-I**, The Hong Kong Polytechnic University, 2017-18 (HK$105,000).
- Development of an Agent-Based Space-Use Prediction Simulation System, **PI**, Start-Up Grant for New Faculty, The Hong Kong Polytechnic University, 2016-19, (HK$500,000).
- User Preference-based Space-use Probability Generation, **PI**, Internal Grant, The Hong Kong Polytechnic University, 2016-18, (HK$500,000).
- Development of a Learning Package for Students to Appreciate the High-Performance Building Development, **Co-I**, DLTC Fund, The Hong Kong Polytechnic University, 2016-17, (HK$116,000).

13 Teaching and learning projects

- XR Industry-Focused Graduate Talent Development Program, **Participating Faculty**, Korea Institute for Advancement of Technology, 2025-2030 (KRW440,000,000).
- Graduate School of Metaverse, **Participating Faculty**, the Institute for Information communication Technology Planning and evaluation, 2022-2027 (KRW550,000,000).
- Creative Content Science Education Research Group (BK21 Four) [Years 2–3], **Co-I**, National Research Foundation, 2020-23 (KRW500,000,000).
- Multi-Intelligent (MI) Space Regeneration Design Talent Program (BK21 Four) [Years 1–2], **Participating Faculty**, National Research Foundation, 2020-27 (KRW2,100,000,000).

- Creative Talent Program for Building Regeneration Design Responding to Future Climate and Demographic Change [Year 6] (BK21 Plus), **Participating Faculty**, National Research Foundation, 2018-2020 (KRW2,500,000,000).
- Development of a Learning Package for Students to Appreciate the High-Performance Building Development, **Co-I**, DLTC Fund, The Hong Kong Polytechnic University, 2016-17, (HK$116,000).

14 Invited talks

35 talks

- "Doing Convergence Research in Architecture", Special Lecture, 18th Academy of Architectural Leadership (AAL), Architectural Institute of Korea, 27 Oct 2026
- "KAIST Research Capabilities and Horizon Europe Collaboration", Austrian Research Promotion Agency (FFG), Vienna, Austria, 10 Sep 2026
- "Introduction to European R&D Innovation Policy Frameworks", Pre-training Session, 2026 Global Benchmarking Program on Regional R&D Policy, Korea Institute of Human Resources Development in Science and Technology (KIRD), Daedeok TechBiz Center, 27 Aug 2026
- "Data-driven Analysis of Human Behavior and Emotion in Indoor Architectural Space", First Wednesday Convergence Forum, KAIST, 10 Sep 2025
- "AI Transformation in Fashion & Beauty", KIFT 2025 – Korea International Fashion Tech Forum, Sungjoo Foundation, 4 Sep 2025
  https://adeko.or.kr/notice/?bmode=view&idx=167442392
- "Analysis of Human Behavior and Responses in Architectural Space Using AI-Based Sensing Data", Architectural Design Committee, AIK Spring Conference, 25 Apr 2025
- "Introduction to ABM and Its Applicability to Space Planning", Korea Environment Institute, 25 Feb 2025
- "Introduction to Metaverse Space Generation Technologies", Kunwon Architects, 28 June 2024
- "The Future of Space", Knowledge Attic, KAIST Center for Gifted Education, 11 Mar 2024
  https://www.youtube.com/watch?v=oW-bZpnWWhs
- "Meta-Museum design and experience", 1st KAIST-NYU Joint workshop on MetaMuseum, 11-12 December 2023
- "The Future Metaverse Space", Department of Architectural Engineering, Chungnam National University, 7 Dec 2023
- "Optimal Design of Metaverse Telemedicine Spaces and Platform Development", 2023 KU Medicine-KAIST Symposium, 20 October 2023
- "Future of Metaverse in AEC industry", The Hong Kong Polytechnic University, 13 October 2023
- "Understanding AI and the Metaverse", Dogok Middle School Parents Association, Gangnam-Seocho Office of Education, 12 September 2023
- "Metaverse Space Design", buildSMART FORUM 2023, buildingSMART Korea, 11 Apr 2023
  https://www.ikld.kr/news/articleView.html?idxno=272359

- "The Metaverse as Future Space", Smart City Graduate Special Lecture, Seoul National University, 11 November 2022
- "Metaverse Space Design", School of Architecture Thursday Lecture, Soongsil University, 10 November 2022
- "Physical and Virtual Space Design", Ewha Womans University, 9 November 2022
- "Metaverse and architecture", The 4th ICCEPM International Seminar, 9 November 2022
- "The Metaverse as Future Space", Yonsei University, 4 November 2022
- "Metaverse Space Design in Everyday Life", Artience Camp, Daejeon Science Festival, 23 October 2022
- "Metaverse and Urban Space", Convergence Research Group, Urban Design Institute of Korea, 23 August 2022.
- "The Future of the Architecture Industry in the Metaverse Era", Tech Advanced, Samsung C&T, 21 April 2022.
- "Smart home for the elderly", @TABLE, Graduate School of Science and Technology Policy, KAIST, 17 January 2022.
- "The effectiveness of photo based IVE spatial experience for hotel room tour", IMPACT 2021, SHTM, the Hong Kong Polytechnic University, 16 December 2021.
- "The Future workplace", 2021 Fall GSCT Colloquium, KAIST, 30 November 2021.
- "Metaverse and architectural design", Perspectives and Prospects on the Metaverse, Daejeon Information & Culture Industry Promotion Agency, 17 November 2021. https://www.youtube.com/watch?v=Rwj17_PioOY&t=177s
- "Virtual architecture and design," Better City and Smarter Life Forum, KAIST, 17 November 2021.
- "Virtual world and architecture" 2nd Technology Seminar, Virtual Convergence Economy Activation Forum, 5 October, 2021. https://www.youtube.com/watch?v=qod7OMRv6QQ&t=320s
- "Human-centered space design," Seoul National University, 24 July, 2020
- "Smart office design: New ways of working," BuildingSmart Forum, 16 April, 2019. https://www.ikld.kr/news/articleView.html?idxno=203071
- "User-centered design and evaluation," Korea Unversity, 4 December, 2018
- "Hong Kong's Built Environment and Mega Projects," Korea University Executive Program in Construction Management, 15 May, 2018
- "User experience design and evaluation," Chosun University, 3 January, 2018.
- "Pre-occupancy Evaluation of Building Design," Yonsei University, 29 December, 2017

15 Media interviews

- [The Future of Space] Best Question Winner, Question Attic, KAIST Center for Gifted Education, 2024
  https://www.youtube.com/watch?v=t26A1_iRF0c&t=37s

- The Future of Space, Question Attic, KAIST Center for Gifted Education, 2024
  https://www.youtube.com/watch?v=qiarLbQeSjM
- Window of KAST (People), Spring Issue, 139 , 32-35, 2023
  https://kast.or.kr/kr/space/publication.php?bbs_data=aWR4PTIwOTkxJnN0YXJ0UGFnZT0wJmxpc3RObz0zNjkmdGFibGU9Y3NfYmJzX2RhdGEmY29kZT1tb2smc2VhcmNoX2l0ZW09JnNlYXJjaF9vcmRlcj0=||&bgu=view&cate=
- [Trend ZIP] Design Jobs in the Metaverse | EP.2, DDP Seoul, Seoul Design Foundation, 2022
  https://www.youtube.com/watch?v=KqoVndmTykk&t=141s
- Physical Barriers Broken Down: Is the Era of Metaverse Hospitals Coming?, Medical Times, 2022
  https://m.medicaltimes.com/News/NewsView.html?ID=1144923
- Planning New Office Spaces with a Location-Tracking Beacon System, Hanyang News, 2019 http://www.newshyu.com/news/articleView.html?idxno=501543

16 University service

- Adjunct Professor, KAIST G-School, 2024-
- Adjunct Professor, Smart City Center, KAIST, 2022-
- Participating Faculty, Post-Metaverse Research Center, KAIST, 2022-
- Participating Faculty, Art & Tech Research Center, KAIST, 2022-
- Participating Faculty, Sumi Jo Performing Arts Research Center, KAIST, 2022-
- Freshman Advisor, KAIST, 2024-26
- Program Director, Culture Technology Minor, KAIST, 2022-24
- Member, KAIST Brand Committee, KAIST, 2022-24
- Chair, CT Branding Committee, KAIST, 2022-24
- Chair, Academic Affairs Committee, Graduate School of Metaverse Convergence, KAIST, 2022-
- Member, New Faculty Search Committee, Graduate School of Culture Technology, KAIST, 2023-25
- Member, Steering Committee, Institute for Humanities and Social Sciences, KAIST, 2021-23
- Member, Curriculum Committee, College of Liberal Arts and Convergence Science, KAIST, 2023-24
- Member, Academic Review Committee, College of Liberal Arts and Convergence Science, KAIST, 2021-23
- Member, Steering Committee, Future Underground City Research Center, KAIST, 2022-23
- Partnership committee member, Hong Kong Polytechnic University, 2015-2018
- New program planning committee member “Building Ecology Engineering”, Hong Kong Polytechnic University, 2017-2018

17 Teaching experience

UNDERGRADUATE

Smart Working Space (https://www.youtube.com/watch?v=gQ0r3KyodVU) | Architectural Modeling I | Digital Design and BIM | FM and Space | FM Research Project | Light, Man, and Environment | Engineering Management

GRADUATE

Metaverse Space Design Research | Virtual Reality and Virtual Worlds | IoT-based Space Design | Low Carbon Building Design | Facility Management Professional Practice | Economics for Facility Management | Indoor Environmental Quality Assessment | Building Carbon Footprint Assessment | Post-Occupancy Evaluation

18 Lab members

30 current

Chun, Sookyung, Research Professor, 07/2022 – present.

Cho, Kyunghyun, Ph.D. student, Principal supervisor, 06/2022 – present.

Eum, Yujeung, Ph.D. student, Principal supervisor, 03/2022 – present.

Lee, Seung Hyun, Ph.D. student, Principal supervisor, 04/2022 – present.

Bae, Seon Ah, Ph.D. student, Principal supervisor, 01/2023 – present.

Myung, Jaekwon, Ph.D. student, Principal supervisor, 01/2026 – present.

Kang, Jisoo, MS student, Principal supervisor, 07/2024 – present.

Jeon, DaHye, MS student, Principal supervisor, 11/2024 – present.

Kim, Yeongjo, MS student, Principal supervisor, 02/2025 – present.

Choo, Shnannon, MS student, Principal supervisor, 02/2025 – present.

Ryu, Yeorim, MS student, Principal supervisor, 09/2025 – present.

Ryu, Najeong, MS student, Principal supervisor, 09/2025 – present.

Lee, Yaejin, MS student, Principal supervisor, 01/2026 – present.

Dohui, Kwon, Intern, Principal supervisor, 03/2026 – present

Gisung, Han, Research Professor, 06/2024 –present

Song, Yoan, Ph.D. student, Principal supervisor, 04/2022 – present.

Choi, Hyeokjin, Ph.D. student, Principal supervisor, 06/2022 – present.

Jang, Jae Young, Ph.D. student, Principal supervisor, 01/2023 – present.

Kim, Bomin, Ph.D. student, Principal supervisor, 09/2025 – present.

Shim, Seoyoung, Ph.D. student, Principal supervisor, 09/2026 – present.

Shin, Jaeyeon, MS student, Principal supervisor, 07/2024 – present.

Chung, Dagyeom, MS student, Principal supervisor, 12/2024 – present.

Kim, Artem, MS student, Principal supervisor, 02/2025 – present.

Seol, Hyo Eun, MS student, Principal supervisor, 05/2025 – present.

Ji, Jiaxian, MS student, Principal supervisor, 09/2025 – present.

Han, Jaewon, MS student, Principal supervisor, 09/2025 – present.

Jiwon, Yang, MS student, Principal supervisor, 03/2026 – present.

Eunseok, Jang, Intern, Principal supervisor, 07/2026 – present

Cherdchanyapong, Kavin, Ph.D. student, Co-supervisor, 09/2025 – present. (GDI program)

Zhang, Shuzhen, Ph.D. student, External co-supervisor, 07/2018 – present. (HK PolyU)

19 Alumni

23 alumni

Chang, Yunseo, MS student, Principal supervisor, 08/2024 – 08/2025

Kim, Hyun Woo, MS student, Principal supervisor, 03/2023 – 08/2025

Lee, Jung Won, MS student, Principal supervisor, 08/2022 – 06/2024

Lee, Hanung, MS student, Principal supervisor, 03/2022– 08/2023

Lee, Chan Kyu, Post-doc, Principal supervisor, 06/2021–04/2022 (Current: Post-doc, Korea Institute of Civil Engineering and Building Technology)

Kim, Kyungeun, MS student, Principal supervisor, 03/2020 – 08/2021

Choi, Heejee, MS student, Principal supervisor,09/2020 – 08/2021

Gassar, Abdo Abdullah Ahmed, Post-doc, Principal supervisor, 03/2020– 03/2021. (Current: Researcher, University of La Rochelle)

Baek, Seunghyo, Postdoctoral Fellow, Principal supervisor, 10/2017– 08/2018. (Current: Assistant Professor, Mokwon University)

Whenzhuo, Li, Ph.D. student, External co-supervisor, 04/2018 – completed.

Ching, Po Chak, MS student, External co-supervisor, 07/2017– completed.

Peng, You, Visiting Ph.D. student, Eindhoven University of Technology, Host supervisor, 08/2016 – 12/2016.

Lee, Junseo, MS student, Principal supervisor, 08/2024 – 08/2025

Jung, Koeun, Research Professor, 02/2024 –08/2024 (Current: Assistant Professor, Chonnam National University)

Abbas, Ali, Post-doc, HK PolyU, Co-supervisor, 12/2021–08/2023 (Current: Assistant Professor, Université Côte d'Azur)

Rhee, Jee Heon, Post-doc, Principal supervisor, 03/2021–06/2022 (Current: Assistant Professor, Gachon University)

Ma, Jae Hoon, MS student, Principal supervisor, 03/2019 – 11/2021 (Current: Assistant Professor, Indiana University Bloomington)

Jo, Taehee, MS student, Principal supervisor, 03/2020 – 08/2021

Yun, Junyoung, MS student, Principal supervisor, 09/2020 – 08/2021

Lee, Byunghee, Postdoctoral Fellow, Principal supervisor, 10/2017– 05/2018. (Current: Researcher, LH Institute)

Kim, Jeonghwan, Postdoctoral Fellow, Principal supervisor, 07/2016– 08/2018. (Current: Assistant Professor, Korea National University of Transportation)

Zhang, Shaojie, Ph.D. student, External co-supervisor, 07/2016 – completed.

Wang, Lyubing, MS student, External co-supervisor, 07/2016 – completed.

20 Honors and certificates

- Member, Young Korean Academy of Science and Technology (YKAST), 2023-
  https://www.dongascience.com/news.php?idx=57523
- Research Innovation Excellence Award, College of Liberal Arts and Convergence Science, KAIST, 2022
- Best paper award, Journal of KIBIM, 2021
- Selected Outstanding Startup, Pre-Startup Package, Ministry of SMEs and Startups, 2021
- Early-Career Researcher Award(TOP 3%, Awarded to two faculty members out of 80), Hanyang University, 2021

- Researcher of the Month (November), Hanyang University, 2020
- Early-Career Researcher Award(TOP 3%, Awarded to two faculty members out of 60), Hanyang University, 2020
- Academic Service Award, Korean Institute of Interior Design, 2019
- Best work award, KIID International exhibition, 2019
- Best paper award, I3CDE, 2019
- Best paper award, KIBIM annual conference, 2019
- Best paper award, KICEM annual conference, 2018
- Dean's award: Outstanding Achievement in Academic Programme Development, 2016
- Introduction to University Teaching, EDC, the Hong Kong Polytechnic University, 2016
- LEED AP O+M (Operation & Maintenance), U.S. Green Building Council, 2010
- Lundgren research awards, University of Cambridge, 2014
- Kettle's Yard travel grant, University of Cambridge, 2014
- Fieldwork research grant, University of Cambridge, 2012-2014
- Erasmus grant, European Commission, 2012-2013
- USGBC Greenbuild scholarship, United States Green Building Council, 2010
- CMAA scholarship, CMAA, North California Chapter, 2010

21 Contribution to professional service

- Associate Editor, Architectural Science Review (A&HCI, Q1, TOP 3%), Taylor & Francis, 2026-
- Member, Architectural Policy Committee, Daejeon Metropolitan City, 2026-2028
- Member, Architecture Committee, Daejeon Metropolitan City, 2026-2028
- Architectural Advisory Committee Member, Korea Industrial Complex Corporation, 2026-2028
- Center Director, AI Urban Research Center, Urban Design Institute of Korea, 2026-
- Member, AI Science and Technology Committee, KAST, 2025-
- Editorial board member, Scientific Reports (SCIE, Q1, Top 15%), Springer Nature, 2024-
- Editorial Board Member, Journal of the Urban Design Institute of Korea, 2026-
- Editorial Board Member, Journal of the Korean Housing Association, 2026-
- Editorial Board Member, Journal of the Korean Institute of Interior Architecture Design, 2026
- Organizing Committee Member, CDE 2021/2023/2026 Summer Conferences, Society for Computational Design and Engineering
- Organizing Committee Member, BuildSmart Conference, buildingSMART Korea, 2022, 2026
- Horizon Europe National Contact Point for Korea, NRF-KERC, 2025-26

- Guest Editor, Special Issue “Revolutionizing Workplace Efficiency with VR/AR, IoT, and AI: Future Office Innovations,” Journal of Corporate Real Estate (ESCI), 2025–2026
- Associate Editor-in-Chief, Journal of the Korean Housing Association (KCI), 2024-2025
- Executive Director, Korea Institute of Ecological Architecture and Environment, 2023-2024
- Evaluation Committee Member, Public Procurement Service (Architectural Planning), 2022-2024
- Evaluation Committee Member, IITP, 2023-
- Editor-in-Chief, Journal of KIBIM (KCI), 2023-24
- Associate Editor, Journal of Computational Design and Engineering (SCIE, Q1, Top 5%) Oxford Univ. Press, 2022-
- Technical committee member, 10th International Conference on Construction Engineering and Project Management (ICCEPM 2024).
- Member, AI Architecture Committee, Architectural Institute of Korea, 2022-2024
- Member, Smart Architecture Committee, Architectural Institute of Korea, 2022-2024
- Member, Interior Architecture Committee, Architectural Institute of Korea, 2022-2024
- Member, Smart+Building Alliance, Ministry of Land, Infrastructure and Transport, 2022-2024
- Preparatory Committee Member, New Content Academy, Korea Creative Content Agency, 2023
- Chair, Convergence Research Committee, Urban Design Institute of Korea, 2022-
- Member, KAST Policy Research Committee, KAST, 2022
- Director, PR Committee, Korea Green Building Council, 2022
- Deputy Editor, Journal of KIBIM (KCI) , 2021-2022
- Director, CG&I Society, Korean Institute of Information Scientists and Engineers, 2023-
- Member, Built Environment Committee, Architectural Institute of Korea, 2020-2022
- Board Member, Society for Computational Design and Engineering, 2020-2025
- Deputy Editor, The BIM Magazine, buildingSMART Korea, 2020-
- Board Member, Korea Institute of Ecological Architecture and Environment, 2019-2022
- Board Member, Korean Institute of Architectural Sustainable Environment and Building Systems, 2020-2025
- Vice Chair, Smart Architecture Committee, Korea Institute of Building Construction, 2019-2022
- Member, Smart Architecture/Construction Special Committee, Architectural Institute of Korea, 2019-2022
- Technical Advisor, Jeju Free International City Development Center, 2019
- Certification Exam Committee Member, Construction Management Professional (CMP), Korea CM Association, 2019
- Guest Editor, Special issue “Health, Safety and Environment (HSE) in the Building and Construction Industry”, Applied Sciences (SCIE, Q2, IF2.90), 2021-2022
- Review Editor, Frontiers in Psychology (SSCI, Q1, IF3.80), 2021-2023

- Guest Editor, Special issue “Sustainable Urban Environment and Building Energy Efficiency”, Sustainability (SCIE, Q2, IF4.00) , 2021-2022
- Guest Editor, Special issue “Advances in Materials, Technologies and Controls for Sustainable Building”, Energies (SCIE, Q3, IF3.20), 2020-2022
- Technical Theme Chair, 8th International Conference on Construction Engineering and Project Management (ICCEPM 2020).
- Scientific Advisory Board at the 3rd International Conference on the Human Factors in Communication of Design track, 24-28 July, 2019, Washington Hilton, Washington D.C., USA.
- Scientific Advisory Board at the 9th International Conference on Applied Human Factors and Ergonomics (AHFE 2018) for the Human Factors in Communication of Design track, 22 - 26 July, 2018, Orlando, Florida, USA.
- Session Chair of Construction Robotics (IV) / Design session at the 34th International symposium on Automation and Robotics in Construction (ISARC 2017), 28 June 28 - 1 July, Taipei, Taiwan.

JOURNAL REVIEWER

Journal of Engineering Design, Elsevier.
Automation in Construction, Elsevier.
Building and Environment, Elsevier.
Journal of Building Engineering, Elsevier.
Displays, Elsevier.
Renewable & Sustainable Energy Reviews, Elsevier.
Advanced Engineering Informatics, Elsevier.
Architectural Engineering and Design Management, Taylor & Francis.
Architectural Science Review, Taylor & Francis.
Building Simulation, Springer Nature.
Scientific Reports, Springer Nature.
Indoor and Built Environment, Sage.
Journal of Management in Engineering, ASCE.
Sensors, MDPI.
Journal of the Society for Computational Design and Engineering
Archives of Design Research, Korean Society of Design Science
Journal of the Korea Institute of Ecological Architecture and Environment
Frontiers of Architectural Research, Elsevier.
Applied Energy, Elsevier.
Resources, Conservation and Recycling, Elsevier.
Library and Information Science Research, Elsevier.
Sustainable Cities and Society, Elsevier.
Journal of Computational Design and Engineering, Elsevier.
Journal of Urban Technology, Taylor & Francis.
Journal of Toxicology and Environmental Health, Taylor & Francis.
International Journal of Human-Computer Interaction, Taylor & Francis.
KSCE Journal of Civil Engineering, Springer Nature.
Journal of Interior Design, Wiley.
IEEE Access, IEEE.
International Journal of Geo-information, MDPI.
Frontiers in Psychology, Frontiers.
Journal of KIBIM
Journal of the Korea Institute of Spatial Design
Journal of the Korean Institute of Culture Architecture

22 Contribution to community service

- Mentor, Youth Science Gifted Mentoring Program, KAST, 2023, 2025, 2026